H Scalability Improvements Corresponding to Our 3 Novelties – A Summary of Computational Complexity and Memory

Our three novelties for enhancing scalability are **slice sampling**, **spatial NNGP prior**, and **sequential updating algorithms**. We first list some important points to note.

- $O \in \mathbb{N}$ is assumed very small; h, k, $p \ll m$; $1 \leq L$ or L_j's $\leq m$.
- Slice sampling is useful when L has to be at least moderately large. Moreover, it resolves intimidating storage concerns regarding θ_{jl}'s, $\boldsymbol{\alpha}_{jl}$'s, and $\boldsymbol{w}_{jl}$'s and thus enables practicable implementation of spatial prediction and clustering when both m and L are large.
- Spatial NNGP prior and sequential updates are helpful for a large m. Sequential updating algorithms can only be adopted when our spatial NNGP prior is specified. Hence, when slice sampling is not adopted, we have 3 models, which we denote as `fullGPfixedL`, `NNGPblockFixedL`, and `NNGPsequenFixedL`.
- When slice sampling is adopted and m is at least moderately large, we may only also specify our spatial NNGP prior with sequential updates for a computationally feasible model, which we shall denote as `NNGPsequenVaryLj`, as multivariate rejection sampling from high-dimensional truncated mO-variate normal distributions of the $\boldsymbol{\alpha}_{jl_j}$'s (Equation (3.10)) is unbearably slow.

Hence, we only implement the four models `fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj` for our simulation experiments. To verify what we have obtained theoretically regarding scalability, we may compare

- `fullGPfixedL` and `NNGPblockFixedL` for effects caused by spatial NNGP prior (without slice sampling, i.e., under our model in Section 2);
- `NNGPblockFixedL` and `NNGPsequenFixedL` for effects caused by sequential updates (without slice sampling, i.e., under our model in Section 2);
- `NNGPsequenFixedL` and `NNGPsequenVaryLj` for effects caused by slice sampling (with our spatial NNGP prior and sequential updating algorithms adopted).

We now elaborate on **theoretical** scalability improvements attributable to each of our three novelties. Computational complexity reduction for certain involved major steps in posterior sampling and spatial prediction, as well as storage requirements alleviation for some concerned key quantities in main model fitting and spatial prediction & clustering, are considered measures of scalability improvements. All computational complexity orders in the subsections below are for only one typical MCMC iteration.

H.1 Flexibility, Computation, and Storage Improvements Brought About by Our Slice Sampling Approach in Section 3 with Spatial NNGP & Sequential Updates

We assume that $O(T) \leq O(m)$. Recall that $1 \leq L$ or L_j's $\leq m$ and that we have assumed $O, h, k, p \ll m$. It is likely restrictive to pre-specify a fixed common $L \in \mathbb{N}$, $L > 1$, as the numbers of spatial mixtures are unknown and may well differ across the k latent factors. Moreover, an at least moderately large L would typically be required to guarantee satisfactory model fitting, resulting in computational burdens and storage issues. Our slice sampling approach

(Section 3) not only brings about desired modeling flexibility and adequacy but also decently addresses the involved scalability issues (Table 5) by ensuring non-increasing posterior samples for L_j's through the MCMC iterations.

Let $N \in \mathbb{N}$ be the number of kept post-burn-in MCMC iterations. We denote the corresponding estimated number(s) of spatial mixture components as $L^{(t_i)}$ or $L_j^{(t_i)}$'s, $i \in \{1, \ldots, N\}$, $j \in \{1, \ldots, k\}$. In Table 5, overall storage orders correspond to all kept MCMC iterations, whereas overall computational complexity orders are only for one typical MCMC iteration.

Domain of Scalability Improvements		**Computational Acceleration**		**Storage Alleviation**	
		Posterior Sampling	**Spatial Prediction** at $r \in \mathbb{N}$ $(r < m)$ new locations		**Spatial Clustering**
Key Parameters / Quantities Involved		$u_j(\boldsymbol{s}_{i,o})$'s / $z_{jl}(\boldsymbol{s}_{i,o})$'s, $\xi_j(\boldsymbol{s}_{i,o})$'s, $\boldsymbol{\alpha}_{jl_j}$'s, θ_{jl_j}'s, $\delta_{1:k}$, κ, ρ	$\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{(m+1):(m+r)})$'s	$\boldsymbol{\alpha}_{jl_j}$'s and θ_{jl_j}'s	$\boldsymbol{w}_{jl_j}$'s
Overall Standard Computational Complexity or Storage Order	**Before**	$\boldsymbol{O\left(kLmO\left\{h^2O+T(p+k)\right\}+mh^3\right)}$	$\boldsymbol{O\left(r\left[h^3+k(L-1)O(h+O)\right]+O^3\right)}$	$\boldsymbol{O\left(mOk\sum_{i=1}^{N}L^{(t_i)}\right)}$	
	After	$\boldsymbol{O\left(mO\sum_{j=1}^{k}L_j\left\{h^2O+T(p+k)\right\}+mh^3\right)}$ $\boldsymbol{+O\left(O^2\sum_{j=1}^{k}\sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}[k^*_{jl_ji}+n^*_{jl_ji}]\right)}$	$\boldsymbol{O\left(r\left[h^3+\sum_{j=1}^{k}(L_j-1)\cdot O(h+O)\right]+O^3\right)}$	$\boldsymbol{O\left(mO\sum_{i=1}^{N}\sum_{j=1}^{k}L_j^{(t_i)}\right)}$	
References	**Section(s)**	Section 3 and Appendices A.2 and A.3	Section 5.1 and Appendix G.2		Section 5.2
	Equations		Equations (5.5) and (G.15)		

Table 5: A theoretical summary table of posterior sampling & spatial prediction computation acceleration and spatial prediction & clustering storage alleviation brought about by our slice sampling approach in Section 3 with our spatial NNGP prior and sequential updates (Section 4). Our simulation experiments comparing the models `NNGPsequenFixedL` and `NNGPsequenVaryLj` should lead to results corresponding well to the table in terms of main model fitting time for the involved parameters and spatial prediction time for $\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{(m+1):(m+r)})$'s.

With our slicing sampling approach (Section 3) adopted on top of our spatial NNGP prior (Section 4) coupled with sequential updates (Section 4.4), rejection sampling from low-dimensional truncated O-variate normal distributions would have adequate acceptance rates and thus small required numbers $n^*_{jl_ji}$'s of draws from $N_O\left(V_{\boldsymbol{s}_i}\boldsymbol{\mu}_{jl_j,\boldsymbol{s}_i}, V_{\boldsymbol{s}_i}\right)$'s (Equation (4.9) in Section 4.4) per sample of $\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_i)$'s. If $O\left(\max_{(j,l_j,i)}\left[k^*_{jl_ji}+n^*_{jl_ji}\right]\right) \leq O(T)$ and $\max_{j\in\{1,\ldots,k\}} L_j \ll m$, then

$$O\left(mO\sum_{j=1}^{k}L_j\left\{h^2O+T(p+k)\right\}+mh^3+O^2\sum_{j=1}^{k}\sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}[k^*_{jl_ji}+n^*_{jl_ji}]\right)=O(mT)\leq O(m^2).$$

Detailed Posterior Sampling Computational Complexity Order Derivations

(a) **Without Slice Sampling (Before)**

$$\begin{aligned}&O\left(\left\{k(L-1)O^2[h^2+h+1]+h^3\right\}m\right)+O\left(\left[k(L-1)h^2O^2+h+O^3\right]m\right)+O(kLmOT(p+k))\\&=\boldsymbol{O\left(kLmO\left\{h^2O+T(p+k)\right\}+mh^3\right)}=O(mT)\leq O(m^2)\text{ if }L\ll m\end{aligned}\tag{H.1}$$

(b) **With Slice Sampling (After)**

$$\begin{aligned}&O\left(\left\{\sum_{j=1}^{k}(L_j-1)\cdot O^2[h^2+h+1]+h^3\right\}m\right)\\&+O\left(mh+O^2\sum_{j=1}^{k}\left[mh^2(L_j-1)+\sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}(k^*_{jl_ji}+n^*_{jl_ji})\right]\right)+O\left(\sum_{j=1}^{k}L_j\cdot mOT(p+k)\right)\end{aligned}$$

$$= \boldsymbol{O}\left(\boldsymbol{mO}\sum_{\boldsymbol{j=1}}^{\boldsymbol{k}}\boldsymbol{L_j}\left\{\boldsymbol{h^2O}+\boldsymbol{T(p+k)}\right\}+\boldsymbol{mh^3}+\boldsymbol{O^2}\sum_{\boldsymbol{j=1}}^{\boldsymbol{k}}\sum_{\boldsymbol{l_j=1}}^{\boldsymbol{L_j-1}}\sum_{\boldsymbol{i=1}}^{\boldsymbol{m}}[\boldsymbol{k^*_{jl_ji}}+\boldsymbol{n^*_{jl_ji}}]\right) \tag{H.2}$$

H.2 Posterior Sampling Computational Acceleration & Storage Alleviation and Faster Spatial Prediction Brought About by Our Spatial NNGP Prior in Section 4

Suppose we have a large m. Recall that $1 \leq L$ or L_j's $\leq m$ and that we have assumed $O, h, k \ll m$. We also assume $r < m$ for spatial prediction at $r \in \mathbb{N}$ new locations $\boldsymbol{s}_{(m+1):(m+r)}$. Note that $[F(\rho)\otimes\kappa]^{-1} = F(\rho)^{-1}\otimes\kappa^{-1}$ and that $\det(F(\rho)\otimes\kappa) = \det(F(\rho))^{O}\cdot\det(\kappa)^{m}$.

Domain of Scalability Improvements		Posterior Sampling		Spatial Prediction
		Acceleration	Storage Alleviation	Acceleration
Parameters Involved		κ and ρ	κ and ρ (and $\boldsymbol{\alpha}_{jl_j}$'s)	$\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{(m+1):(m+r)})\vert\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{1:m}), \kappa, \rho$'s for all (j, l_j)
Key Quantities Concerned		$F(\rho)^{-1}$ and $\det(F(\rho))$	$F(\rho)$ and $F(\rho)^{-1}$	conditional normal variances & their Cholesky factors and mean vectors
Standard Computational Complexity or Storage Order for the Key Quantities	Before	$O(m^3)$	$\boldsymbol{O(m^2)}$	$O(m^3)+O(r^3)+O(O^3)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot rmO\right)$
	After	$O(mh^3)$	$\boldsymbol{O(mh^2)}$	$O(rh^3)+O(r)+O(O^3)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot r\cdot hO\right)$
Overall Standard Order of Computational Complexity Without Slice Sampling	Before	$\boldsymbol{O(m^3)}$		$\boldsymbol{O(m^3)}$
	After	$\boldsymbol{O\left(\left\{k(L-1)O^2[h^2+h+1]+h^3\right\}m\right)}$		$\boldsymbol{O\left(r\left[h^3+k(L-1)O(h+O)\right]+O^3\right)}$
Overall Standard Order of Computational Complexity With Slice Sampling	Before	$\boldsymbol{O(m^3)}$		$\boldsymbol{O(m^3)}$
	After	$\boldsymbol{O\left(\left\{\sum_{j=1}^{k}(L_j-1)\cdot O^2[h^2+h+1]+h^3\right\}m\right)}$		$\boldsymbol{O\left(r\left[h^3+\sum_{j=1}^{k}(L_j-1)\cdot O(h+O)\right]+O^3\right)}$
References	Section(s)	Section 4.3	Sections 4.2 and 4.3	Section 5.1 and Appendix G.2.2
	Equations	Equations (A.18), (A.19), (A.34) and (A.35)		Equations (5.4), (5.5), (G.14) and (G.15)

Table 6: A theoretical summary table of posterior sampling computation acceleration & storage alleviation and spatial prediction acceleration brought about by our spatial NNGP prior in Section 4 with or without adopting slice sampling (Section 3). Our simulation experiments comparing the models `fullGPfixedL` and `NNGPblockFixedL` should lead to results corresponding well to the table in terms of main model fitting time for κ and ρ and spatial prediction time for $\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{(m+1):(m+r)})$'s.

Detailed Derivations Regarding the Overall Computational Complexity Orders

1. Computational Complexity for the Posterior Sampling Steps of ρ and $\kappa_{O\times O}$

(a) The Full Spatial GP Prior Adopted (Before)

Without Slice Sampling (available for simulation comparisons)

$$\left[O(m^3)+O\left(k(L-1)(Om)^2\right)\right]+O\left(k(L-1)\cdot Om^2\right)=\boldsymbol{O(m^3)} \tag{H.3}$$

With Slice Sampling

$$\left[O(m^3)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot(Om)^2\right)\right]+O\left(\sum_{j=1}^{k}(L_j-1)\cdot Om^2\right)=\boldsymbol{O(m^3)} \tag{H.4}$$

(b) Our Spatial NNGP Prior in Section 4 Adopted (After)

Without Slice Sampling (available for simulation comparisons)

$$\begin{aligned}&\left[\left\{O(mh^3)+O(O^3)+O\left(m(h+1)^2\right)+O(m)\right\}+O\left(k(L-1)m[h^2+h+1]O^2\right)\right]\\&+O\left(k(L-1)\cdot Om[h^2+h+1]\right)\\=&\boldsymbol{O\left(\left\{k(L-1)O^2[h^2+h+1]+h^3\right\}m\right)}=O(m)\text{ if }L\ll m\end{aligned} \tag{H.5}$$

With Slice Sampling

$$\left[\left\{O\left(mh^3\right)+O\left(O^3\right)+O\left(m(h+1)^2\right)+O(m)\right\}+O\left(\sum_{j=1}^{k}(L_j-1)\cdot m[h^2+h+1]O^2\right)\right]$$
$$+O\left(\sum_{j=1}^{k}(L_j-1)\cdot Om[h^2+h+1]\right)$$
$$=\boldsymbol{O}\left(\left\{\sum_{j=1}^{k}(\boldsymbol{L_j-1})\cdot\boldsymbol{O^2}[\boldsymbol{h^2+h+1}]+\boldsymbol{h^3}\right\}\boldsymbol{m}\right)=O(m)\text{ if }\max_{j\in\{1,\ldots,k\}}L_j\ll m \tag{H.6}$$

2. Computational Complexity for the Spatial Prediction Step Obtaining $\hat{\alpha}_{jl_j}\left(\boldsymbol{s}_{(m+1):(m+r)}\right)$ for all (j,l_j)

(a) The Full Spatial GP Prior Adopted (Before)

Without Slice Sampling (available for simulation comparisons)

$$O\left(m^3\right)+O\left(r^3\right)+O\left(O^3\right)+O(k(L-1)\cdot rmO)+O\left(k(L-1)\cdot(rO)^2\right)=\boldsymbol{O}\left(\boldsymbol{m^3}\right) \tag{H.7}$$

With Slice Sampling

$$O\left(m^3\right)+O\left(r^3\right)+O\left(O^3\right)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot rmO\right)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot(rO)^2\right)=\boldsymbol{O}\left(\boldsymbol{m^3}\right) \tag{H.8}$$

(b) Our Spatial NNGP Prior in Section 4 Adopted (After)

Without Slice Sampling (available for simulation comparisons)

$$O\left(rh^3\right)+O(r)+O\left(O^3\right)+O(k(L-1)\cdot r\cdot hO)+O\left(k(L-1)\cdot r\cdot O^2\right)=\boldsymbol{O}\left(\boldsymbol{r}\left[\boldsymbol{h^3}+\boldsymbol{k(L-1)O(h+O)}\right]+\boldsymbol{O^3}\right) \tag{H.9}$$

With Slice Sampling

$$O\left(rh^3\right)+O(r)+O\left(O^3\right)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot r\cdot hO\right)+O\left(\sum_{j=1}^{k}(L_j-1)\cdot r\cdot O^2\right)$$
$$=\boldsymbol{O}\left(\boldsymbol{r}\left[\boldsymbol{h^3}+\sum_{j=1}^{k}(\boldsymbol{L_j-1})\cdot\boldsymbol{O(h+O)}\right]+\boldsymbol{O^3}\right) \tag{H.10}$$

H.3 Posterior Sampling Computational Acceleration Brought About by Our Sequential Updating Methods in Section 4.4 (with spatial NNGP)

We have calculated $B_{\boldsymbol{s}_i}$'s and $F_{\boldsymbol{s}_i}$'s in the Gibbs sampler step updating ρ. Hence, sequentially sampling the elements $\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_i), i\in\{1,\ldots,m\}$, of $\boldsymbol{\alpha}_{jl_j}$ one by one according to Equation (4.8) incurs a computational complexity of $O\left(m(O^3+h)\right)+O\left(mk(L-1)h^2O^2\right)=O\left([k(L-1)h^2O^2+h+O^3]m\right)=O(m)$ if $L\ll m$. If we adopt our model framework in Section 3 and proceed with Equation (4.9) instead, then the corresponding computational complexity updating all $\boldsymbol{\alpha}_{jl_j}\left(\boldsymbol{s}_i\right)$'s is

$$O(O^3)+O(mh)+O\left(m\sum_{j=1}^{k}(L_j-1)\cdot h^2O^2\right)+O\left(\sum_{j=1}^{k}\left[mO+\sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}(k^*_{jl_ji}+n^*_{jl_ji})\cdot O^2\right]\right)$$

$$= O\left(mh + O^2 \sum_{j=1}^{k} \left[mh^2(L_j - 1) + \sum_{l_j=1}^{L_j-1} \sum_{i=1}^{m} (k^*_{jl_ji} + n^*_{jl_ji})\right]\right),$$

where $k^*_{jl_ji}$'s denote the number of flops required for case discussion and upper and lower bounds calculation. $n^*_{jl_ji}$'s represent numbers of draws from $N_O\left(V_{\boldsymbol{s}_i}\boldsymbol{\mu}_{jl_j,\boldsymbol{s}_i}, V_{\boldsymbol{s}_i}\right)$ required per sample. Sequentially updating $\boldsymbol{\alpha}_{jl_j}$'s as above successfully replaces the unbearably slow rejection sampling from a high-dimensional truncated mO-variate normal distribution by computationally achievable low-dimensional truncated O-variate normal samplings. As the number of observation types $O \in \mathbb{N}$ is typically very small and often equals 1, the truncated O-variate normal samplings (Equation (4.9)) concerned with sequential updating would have adequate acceptance rates with small $n^*_{jl_ji}$'s. If the largest number of mixture components $\max_{j\in\{1,\ldots,k\}} L_j$ is of constant order or increases only logarithmically with m as in Dirichlet processes, then given $k, h, O \ll m$ and $\sum_{j=1}^{k} L_j \leq k \cdot \max_{j\in\{1,\ldots,k\}} L_j$, sequential updates in our Gibbs samplers can reduce the posterior sampling computational complexity for updating all $\boldsymbol{\alpha}_{jl_j}$'s in each MCMC iteration from $O(m^3) + O\left(\sum_{j=1}^{k}\sum_{l_j=1}^{L_j-1} n_{jl_j} \cdot m^2\right)$, which is likely greater than $O(m^3)$, to about linear in m (Table 7 in Appendix H.3).

Suppose we have a large m. Recall that $1 \leq L$ or L_j's $\leq m$ and that we have assumed $O, h, k \ll m$. When slice sampling (Section 3) is adopted without our sequential updating method, rejection sampling from a high-dimensional truncated $mO-$variate normal distribution (Equation (4.7)) gets unbearably slow for feasible computation when m gets large, as a very small acceptance rate leads to a huge number $n_{jl_j} \in \mathbb{N}$ of draws from $N_{mO}\left(\mathbf{0}, \tilde{F}(\rho) \otimes \kappa\right)$ required per sample of $\boldsymbol{\alpha}_{jl_j}$ for each (j, l_j). When m is large, we typically have $n_{jl_j}(m) \gg m$ for any (j, l_j). With our spatial NNGP prior coupled with sequential updates, rejection sampling from low-dimensional truncated O-variate normal distributions would have adequate acceptance rates and thus small required numbers $n^*_{jl_ji}$'s of draws from $N_O\left(V_{\boldsymbol{s}_i}\boldsymbol{\mu}_{jl_j,\boldsymbol{s}_i}, V_{\boldsymbol{s}_i}\right)$'s (Equation (4.9) in Section 4.4) per sample of $\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_i)$'s. If $\max_{j\in\{1,\ldots,k\}} L_j \ll m$ and $\max_{(j,l_j,i)} n^*_{jl_ji} \ll m$, then

$$O\left(mh + O^2 \sum_{j=1}^{k} \left[mh^2(L_j - 1) + \sum_{l_j=1}^{L_j-1} \sum_{i=1}^{m} (k^*_{jl_ji} + n^*_{jl_ji})\right]\right) = O(m).$$

Place of Scalability Improvements		Computational Acceleration for the Posterior Sampling Step of $\boldsymbol{\alpha}_{jl_j}$'s for all (j, l_j)	
		Without Slice Sampling (available for simulation comparisons)	With Slice Sampling
Key Quantities Concerned		$\left[I_{mO} + \tilde{F}(\rho)^{-1} \otimes \kappa^{-1}\right]^{-1}$, $\mathrm{Chol}\left(\left[I_{mO} + \tilde{F}(\rho)^{-1} \otimes \kappa^{-1}\right]^{-1}\right)$	$\mathrm{Chol}\left(\tilde{F}(\rho) \otimes \kappa\right)$ and n_{jl_j}'s for $mO-$variate rejection sampling
Overall Order of Standard Computational Complexity	Before	$O(m^3) + O(k(L-1) \cdot (mO)^2) = \boldsymbol{O(m^3)}$	$\boldsymbol{O(m^3) + O\left(\sum_{j=1}^{k}\sum_{l_j=1}^{L_j-1} n_{jl_j} \cdot m^2\right) > O(m^3)}$
	After	$\boldsymbol{O([k(L-1)h^2O^2 + h + O^3]m)} = O(m)$ if $L \ll m$	$\boldsymbol{O\left(mh + O^2\sum_{j=1}^{k}\left[mh^2(L_j-1) + \sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}(k^*_{jl_ji} + n^*_{jl_ji})\right]\right)}$
References	Section(s)	Section 4.4	
	Equations	Equations (4.6) and (4.8)	Equations (4.7) and (4.9)

Table 7: A theoretical summary table of posterior sampling computation acceleration further brought about by our sequential updating methods in Section 4.4 on top of imposing our spatial NNGP prior (Section 4) with or without adopting slice sampling (Section 3). Our simulation experiments comparing the models `NNGPblockFixedL` and `NNGPsequenFixedL` should lead to results corresponding well to the table in terms of main model fitting time for $\boldsymbol{\alpha}_{jl_j}$'s.

H.4 An Overall Scalability Improvements Summary for Our Three Novelties

Suppose we have a large m. Recall that $1 \leq L$ or L_j's $\leq m$ and that we have assumed $O, h, k, p \ll m$. We also assume $r < m$ for spatial prediction at $r \in \mathbb{N}$ new locations $\boldsymbol{s}_{(m+1):(m+r)}$. Let $N \in \mathbb{N}$ be the number of kept post-burn-in MCMC iterations. We denote the corresponding estimated number(s) of spatial mixture components as $L^{(t_i)}$ or $L_j^{(t_i)}$'s, $j \in \{1, \ldots, k\}$, $i \in \{1, \ldots, N\}$. In Table 8, storage requirements for spatial prediction and clustering correspond to all kept MCMC iterations, whereas computational complexity orders are only for one typical MCMC iteration.

Simulation Model		`fullGPfixedL`	`NNGPblockFixedL`	`NNGPsequenFixedL`	`NNGPsequenVaryLj`
Spatial NNGP Prior		No	Yes	Yes	Yes
Sequential Updating Methods		No	No	Yes	Yes
Slice Sampling Approach		No	No	No	Yes
Spatial Prediction Speed for $\hat{\boldsymbol{\alpha}}_{jl_j}(\boldsymbol{s}_{(m+1):(m+r)})\vert\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{1:m}), \kappa, \rho$**'s**		$\boldsymbol{O(m^3)}$	$\boldsymbol{O\left(r\left[h^3 + k(L-1)O(h+O)\right] + O^3\right)}$		$\boldsymbol{O\left(r\left[h^3 + \sum_{j=1}^{k}(L_j - 1)\cdot O(h+O)\right] + O^3\right)}$
Posterior Sampling Speed for	ρ and $\kappa_{O\times O}$	$\boldsymbol{O(m^3)}$	$\boldsymbol{O\left(\left\{k(L-1)O^2[h^2+h+1]+h^3\right\}m\right)}$		$\boldsymbol{O\left(\left\{\sum_{j=1}^{k}(L_j-1)\cdot O^2[h^2+h+1]+h^3\right\}m\right)}$
	$\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_{1:m})$'s	$\boldsymbol{O(m^3)}$		$\boldsymbol{O([k(L-1)h^2O^2+h+O^3]m)}$	$\boldsymbol{O\left(mh + O^2\sum_{j=1}^{k}\left[mh^2(L_j-1)+\sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}(k^*_{jl_ji}+n^*_{jl_ji})\right]\right)}$
	$z_{jl}(\boldsymbol{s}_{i,o})$'s or $u_j(\boldsymbol{s}_{i,o})$'s	$\boldsymbol{O(kLmO)}$			$\boldsymbol{O(kmO)}$
	$\delta_{1:k}$, θ_{jl_j}'s, and $\xi_j(\boldsymbol{s}_{i,o})$'s	$\boldsymbol{O(kLmOT(p+k))}$ (absolute highest order from $\xi_j(\boldsymbol{s}_{i,o})$'s)			$\boldsymbol{O\left(\sum_{j=1}^{k}L_j\cdot mOT(p+k)\right)}$ (absolute highest order from $\xi_j(\boldsymbol{s}_{i,o})$'s)
Storage Requirements for	$F(\rho)$ and $F(\rho)^{-1}$	$\boldsymbol{O(m^2)}$	$\boldsymbol{O\left(mh^2\right)}$		
	Spatial Prediction	$\boldsymbol{O\left(mOk\sum_{i=1}^{N}L^{(t_i)}\right)}$			$\boldsymbol{O\left(mO\sum_{i=1}^{N}\sum_{j=1}^{k}L_j^{(t_i)}\right)}$
	Spatial Clustering				

Table 8: An overall theoretical summary table of scalability improvements brought about by our 3 novelties. For each of the four models implemented in our simulation experiments, we list its computational complexity orders for involved Gibbs sampler steps in posterior sampling and a major spatial prediction procedure. We also present memory requirements for two key quantities and spatial prediction & clustering under our four models.

With our slice sampling approach (Section 3) adopted on top of our spatial NNGP prior (Section 4) coupled with sequential updates (Section 4.4), rejection sampling from low-dimensional truncated O-variate normal distributions would have adequate acceptance rates and thus small required numbers $n^*_{jl_ji}$'s of draws from $N_O\left(V_{\boldsymbol{s}_i}\boldsymbol{\mu}_{jl_j,\boldsymbol{s}_i}, V_{\boldsymbol{s}_i}\right)$'s (Equation (4.9) in Section 4.4) per sample of $\boldsymbol{\alpha}_{jl_j}(\boldsymbol{s}_i)$'s. If $\max_{j\in\{1,\ldots,k\}} L_j \ll m$ and $\max_{(j,l_j,i)} n^*_{jl_ji} \ll m$, then

$$O\left(mh + O^2\sum_{j=1}^{k}\left[mh^2(L_j-1) + \sum_{l_j=1}^{L_j-1}\sum_{i=1}^{m}(k^*_{jl_ji}+n^*_{jl_ji})\right]\right) = O(m).$$

I Extension to Three Non-Gaussian Observed Data Types

Earlier on, we have only considered Gaussian observed data under all our three modeling frameworks (Sections 2 and 3). As we shall see in this section, our models can be conveniently extended to accommodate three non-normal observed data types – `'tobit'`, `'probit'`, and `'binomial'`, which can all still lead to Gaussian kernels. All we need to add is an extra `SampleY()` step sampling transformed normal response variables from our original observed non-normal data and based on our current posterior parameter estimates at the start of each